

Laplace Transform Notes

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1. Definition

Definitions

The *Laplace transform* is an operation that takes a function of a real variable (usually t) as input and outputs a function of a complex variable s . If $f(t)$ is our original function, $F(s)$ is its transform.

$$\mathcal{L}\{f(t)\}(s) = F(s) \stackrel{\text{def}}{=} \int_0^{\infty} f(t)e^{-st} dt$$

There is a slight caveat. The above integral may not converge. We will ignore this for now.

Example 1

Calculate $\mathcal{L}\{e^{at}\}$.

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{at} e^{-st} dt \\ &= \frac{1}{s-a}\end{aligned}$$

This is not exactly correct. The integral converges only when $\Re(s) > a$. But we will wave our hands and simply say

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a} \text{ for } s \neq a$$

2. Properties of the Laplace Transform

2.1. Linearity

Linearity

The Laplace transform is linear!

$$\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}$$

which follows from the linearity of integrals.

2.2. Existence

Existence Theorem

A function f is of *exponential order* if $|f(t)| \leq Me^{ct}$ for some positive M and c .

If f is continuous and of exponential order with constant c , then $\mathcal{L}\{f\}$ exists for all $s > c$.

Proof follows by comparison.

2.3. Unique and Invertible

Existence Theorem

Two integrable functions have the same Laplace transform if they differ on a set of Lebesgue measure zero. For example, if two functions are different at only finitely many points, they have the same Laplace transform.

We will define an inverse operation

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

3. Some Useful Laplace Transforms

Table of transforms

Assume f is sufficiently nice.

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \text{ for } s \neq a$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$$

$$\mathcal{L}\{\ln t\} = \frac{\ln s + \gamma}{s}$$

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s} \text{ where } u \text{ is the Heaviside step function}$$

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

$$\mathcal{L}\{u(t-a)f(t-a)\} = e^{-as}F(s)$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sp}} \int_0^p e^{-s\tau} f(\tau) d\tau \text{ where } p \text{ is the period}$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$$

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \text{ (the shift property)}$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s') ds'$$

4. Misc Notes

$$\mathcal{L}\{f'\} \rightarrow \text{use by parts}$$

$$\mathcal{L}\left\{\int_0^t \right\} \rightarrow \text{change order of integration, otherwise convolution}$$

$$(f * g)(t) \stackrel{\text{def}}{=} \int_{-\infty}^\infty f(\tau)g(t-\tau)d\tau$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

Set $G = 1$ (the step function) and the result follows.

This holds for nested integrals!

$$\mathcal{L}\left\{\int^n\right\} = \frac{F(s)}{s^n}$$

This is true by Cauchy's repeated integration formula, and then convolution theorem.

Feynman's trick!

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

This generalizes into $t^n f(t)$.

We want to prove $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s')ds'$. We will use the same Feynman's trick, again!

$$\begin{aligned} Q(s') &= \int_0^\infty e^{-st} \frac{f(t)}{t} dt \\ \frac{dQ}{ds'} &= - \int_0^\infty e^{-st} f(t) dt = F(s) \\ \int dQ &= - \int F(s') ds' \end{aligned}$$

Now we want to set appropriate bounds for the above integral. One of the bounds should be s , the variable we want. What should the other one be? 0 doesn't work, the next thing we try is ∞ , since that makes the exponential term zero.

$$\begin{aligned} \int_s^\infty dQ &= - \int_s^\infty F(s') ds' \\ Q(s) &= \int_s^\infty F(s') ds' \end{aligned}$$

Now let's try $\mathcal{L}\{\ln t\}$.

$$\begin{aligned} \int_0^\infty e^{-st} \ln t dt &= \lim_{n \rightarrow \infty} \int_0^{\frac{n}{s}} \left(1 - \frac{st}{n}\right)^n \ln t dt \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \int_0^1 x^n \ln\left(\frac{n(1-x)}{s}\right) dx \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(\int_0^1 x^n \ln(1-x) dx + \int_0^1 x^n \ln\left(\frac{n}{s}\right) dx \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \int_0^1 x^n \sum_{j=1}^\infty \frac{x^j}{j} dx + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \sum_{j=1}^\infty \int_0^1 \frac{x^{j+n}}{j} dx + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \sum_{j=1}^\infty \frac{1}{j(j+n+1)} + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \end{aligned}$$

Now let us look at that infinite sum

$$\begin{aligned}
\sum_{j=1}^{\infty} \frac{1}{j(j+n+1)} &= \frac{1}{n+1} \sum_{j=1}^{\infty} \left(\frac{1}{j} - \frac{1}{j+n+1} \right) \\
&= \frac{1}{n+1} \sum_{j=1}^{n+1} \frac{1}{j} \\
&= \frac{H_{n+1}}{n+1}
\end{aligned}$$

For large n , $H_n \sim \ln(n) + \gamma$. Using this,

$$\begin{aligned}
\mathcal{L}\{\ln t\} &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(-\sum_{j=1}^{\infty} \frac{1}{j(j+n+1)} + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\
&= \lim_{n \rightarrow \infty} \frac{n}{s} \frac{\ln\left(\frac{n}{s}\right) - \ln(n) - \gamma}{n+1} \\
&= \boxed{\frac{\ln s + \gamma}{s}}
\end{aligned}$$

Alternate method:

$$\begin{aligned}
\mathcal{L}\{t^n\} &= \frac{\Gamma(n+1)}{s^{n+1}} \\
\frac{d}{dn} \int_0^{\infty} e^{-st} t^n dt &= \frac{d}{dn} \frac{\Gamma(n+1)}{s^{n+1}} \\
\frac{d}{dn} \int_0^{\infty} e^{-st} t^n \ln t dt &= \frac{s^{n+1} \Gamma'(n+1) - \Gamma(n+1) s^{n+1} \ln s}{s^{2n+2}}
\end{aligned}$$

Set $n = 0$ and compute to get the result. However $\Gamma'(1)$ is a little tricky to get, and needs the same series expansion as above.

5. Motivation

How would we come up with the definition of the Laplace transform? We will look into the history of its inventor, Pierre-Simon Laplace. Laplace was studying celestial mechanics, and required probability theory for some of his calculations. In order to count things, he had the idea of *generating functions*. Consider a sequence a_n . The generating function of a_n is the following power series, denoted $A(x)$:

$$A(x) = a_0 + a_1 x + a_2 x^2 + \dots = \sum_{n=0}^{\infty} a_n x^n$$

Note that we “transform” a function of n to one of x .

Laplace was also studying difference equations and differential equations at the time, and noticed that difference equations were discrete version of differential equations. So it is perhaps a natural question to ask: is there a continuous version of a generating function? Indeed there is, it is the Laplace transform. Firstly replace a_n with $a(n)$.

$$\begin{aligned}
A(x) &= \sum_{n=0}^{\infty} a(n)x^n \\
&= \sum_{n=0}^{\infty} a(n)e^{n \ln x} \\
&\stackrel{s = -\ln x}{=} \sum_{n=0}^{\infty} a(n)e^{-ns} \Delta n
\end{aligned}$$

First we use e instead of x as a base. Since $\ln x$ is negative, we replace it by a new variable $-s$. For convergence we would like $0 < x < 1 \implies 0 < s < \infty$. Also note that Δn here is equal to one, since this is a discrete sequence. Now instead consider a function $a(t)$ and let $\Delta t \rightarrow 0$.

$$A(s) = \int_0^{\infty} a(t)e^{-st} dt$$

which is the Laplace transform.

Now we will see a few properties of generating functions and their Laplace extensions.

Sequence $\longleftrightarrow$ Generating Function

$$(a_0, a_1, \dots) \longleftrightarrow A(x)$$

$$(a_0 + b_0, a_1 + b_1, \dots) \longleftrightarrow A(x) + B(x)$$

$$(0, a_0, a_1, \dots) \longleftrightarrow xA(x)$$

$$(0, 0, \dots, n \text{ zeros}, a_0, a_1, \dots) \longleftrightarrow x^n A(x)$$

$$(1a_1, 2a_2, 3a_3, \dots) \longleftrightarrow A'(x)$$

The second property is linearity.

The fourth property is shifting. The equivalent laplace property can be seen by letting $\ln x = -s$ or $x = e^{-s}$ we get $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$.

The fifth property states that $\mathcal{L}\{tf(t)\} = -F'(s)$. When taking the derivative there's a negative sign because $x = e^{-s}$.

Derivative in the discrete world are instead finite differences. So $\frac{df}{dt}$ would correspond to $\frac{\Delta a}{\Delta n} = a_{n+1} - a_n$. So the sequence

$$(a_1 - a_0, a_2 - a_1, \dots) \longleftrightarrow \frac{(1-x)A(x) - a_0}{x}$$

...incomplete. Similarish $x/(1-x)$ factor comes when integrating (ie summing)

Another way to motivate the convolution theorem is to first see it in generating functions. Let a_n and b_n be two sequences. Then let $C(x) = A(x)B(x)$. Then the coefficients of c_n are given by

$$c_n = \sum a_k b_{n-k} = a * b.$$

Which is a convolution! And similarly $\mathcal{L}\{f * g\} = F(s)G(s)$.

An example of a difference equation is $a_n - a_{n-1} = a_{n-2}$, with $a_0 = 0$ and $a_1 = 1$. A way of solving this equation is to let $A(x)$ be the generating function of a_n . So

$$\begin{aligned}
a_n - a_{n-1} &= a_{n-2} \\
a_n x^n - a_{n-1} x^n &= a_{n-2} x^n \\
\sum_{n=2}^{\infty} a_n x^n - \sum_{n=2}^{\infty} a_{n-1} x^n &= \sum_{n=2}^{\infty} a_{n-2} x^n \\
(A(x) - a_0 - a_1 x) - (xA(x)) &= x^2 A(x) \\
A(x) &= \frac{x}{1 - x - x^2}
\end{aligned}$$

Now by analyzing $A(x)$, we can find the coefficients of x^n which is a_n by definition.

to-dos

heaviside's method (D operator) ref: Princeton companion to math finish up gen fuction argument
more to do: <https://3x10e8.wordpress.com/wp-content/uploads/2014/01/laplace-transform-motivation.pdf>

"laplace transform diagonalizes the derivative operator"??